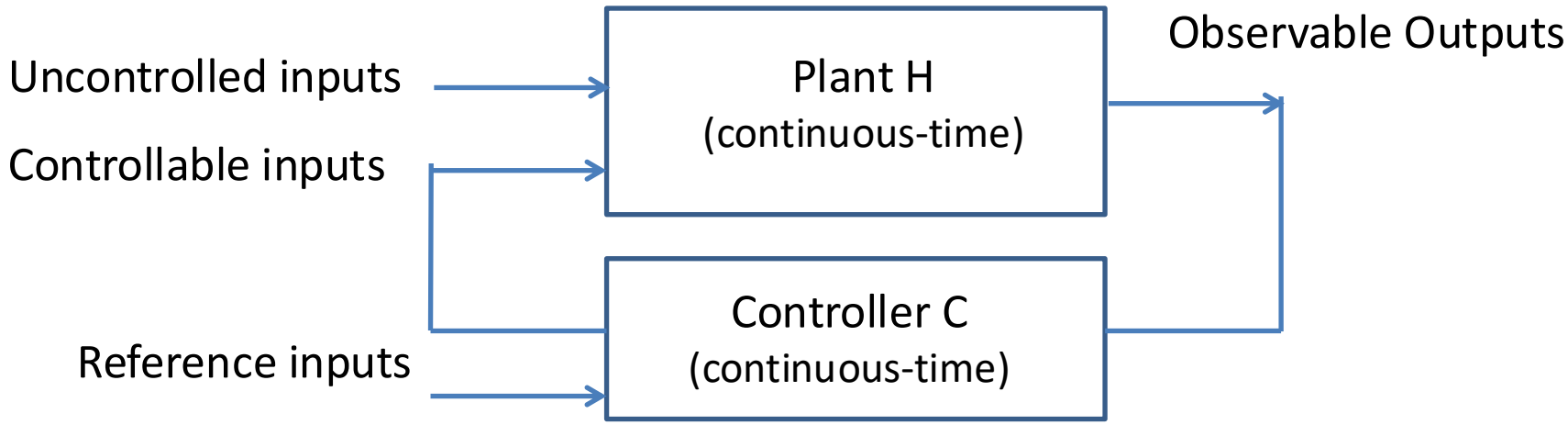
CS:5980
Foundations of Embedded Systems

Dynamical Systems

Part III

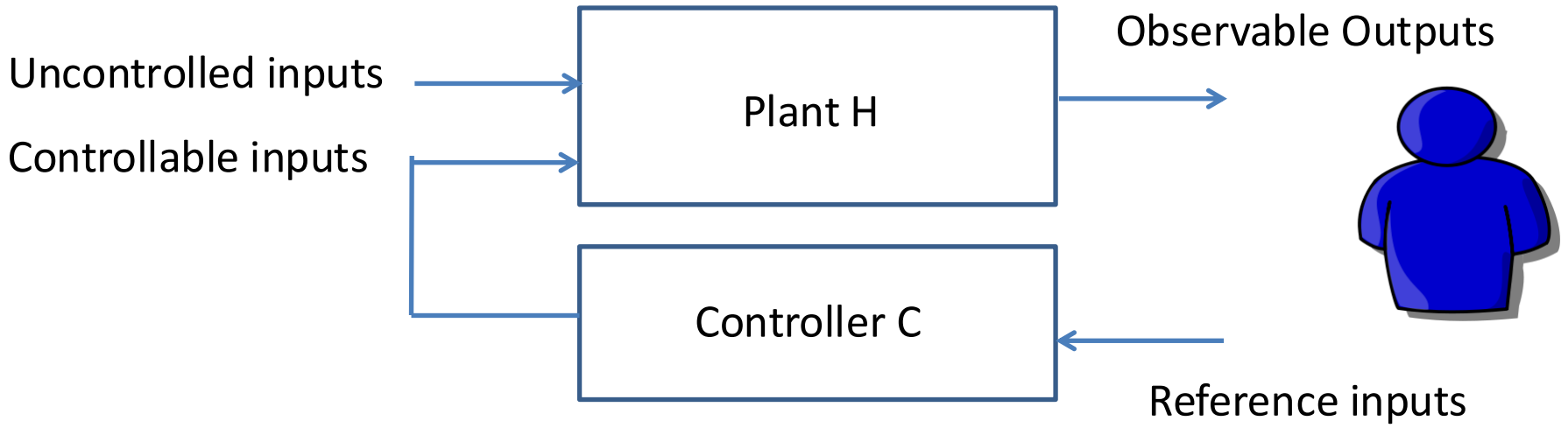
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Control Design Problem



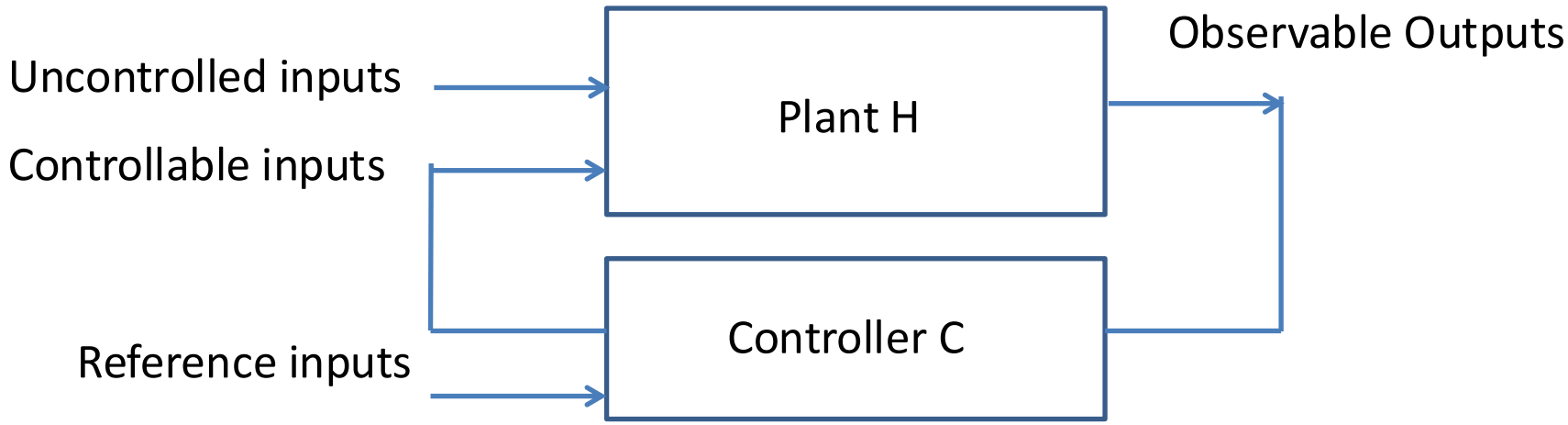
- Design a controller C so that the composed system C || H is stable
- Reference inputs are high-level commands supplied by users (e.g., desired speed of the car, temperature in the room)
- Controller should satisfy additional safety/liveness requirements (e.g., car speed eventually comes close to desired cruising speed)

Open Loop Controller



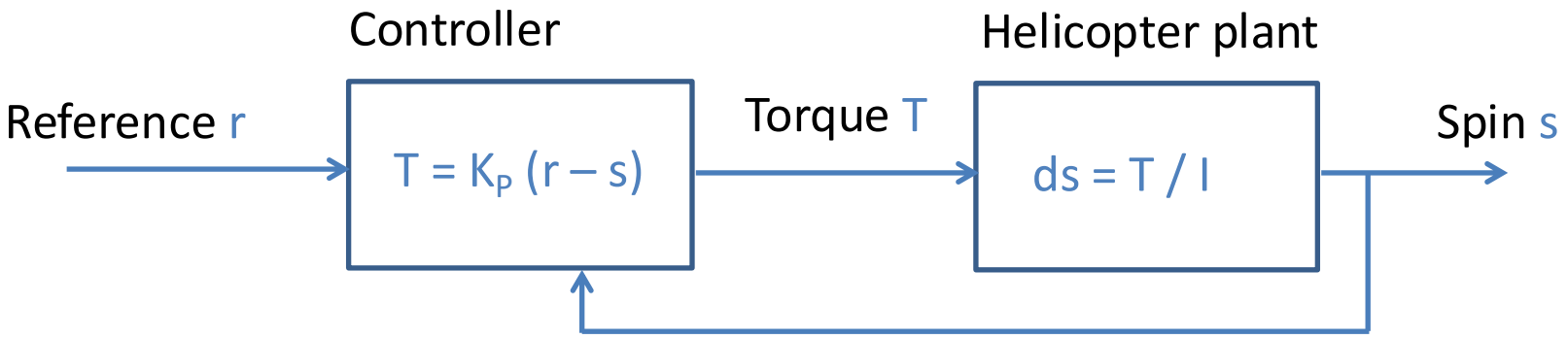
- Plant outputs not fed to the controller
  - **Benefit:** Sensors not needed (less expensive)
- Controller simply maps reference inputs to controllable inputs
  - Knowledge of plant dynamics hard-coded in this algorithm
- Human intervention typically necessary to maintain acceptable performance

Feedback Controller



- Controller adjusts controllable inputs in response to outputs
  - Can respond better to variations in disturbances
  - Performance depends on how well outputs can be measured
- Two control design techniques:
  1. Mathematical, based on theory of linear systems
  2. PID controllers, widely used in practice

Feedback Controller for Helicopter Model

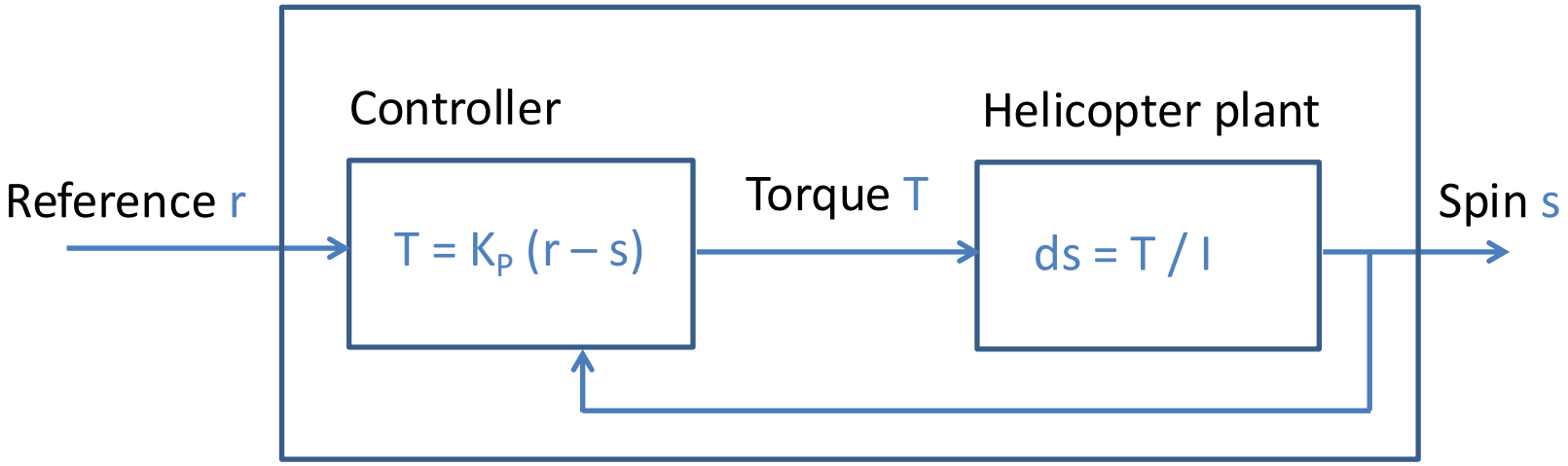


Design controller so that composed system is stable

- *Error* e = (r – s): difference in desired value r and observed output s
- *Proportional controller*: output T is proportional to error e
- *Proportional gain*: Constant K_P

Note: the direction of torque changes with sign of the error

Stabilizing Controller for Helicopter Model

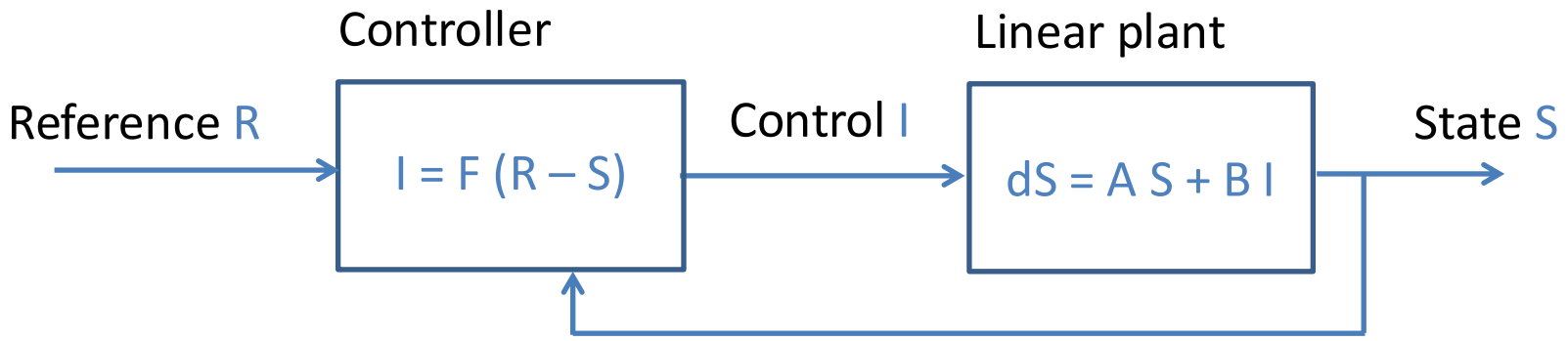


- Dynamics of the composed system: $ds = K_P (r - s) / I$
- When is this system asymptotically stable? BIBO stable?
  - When the coefficient $-K_P / I$ is negative

Control design: choose a positive gain constant K_P

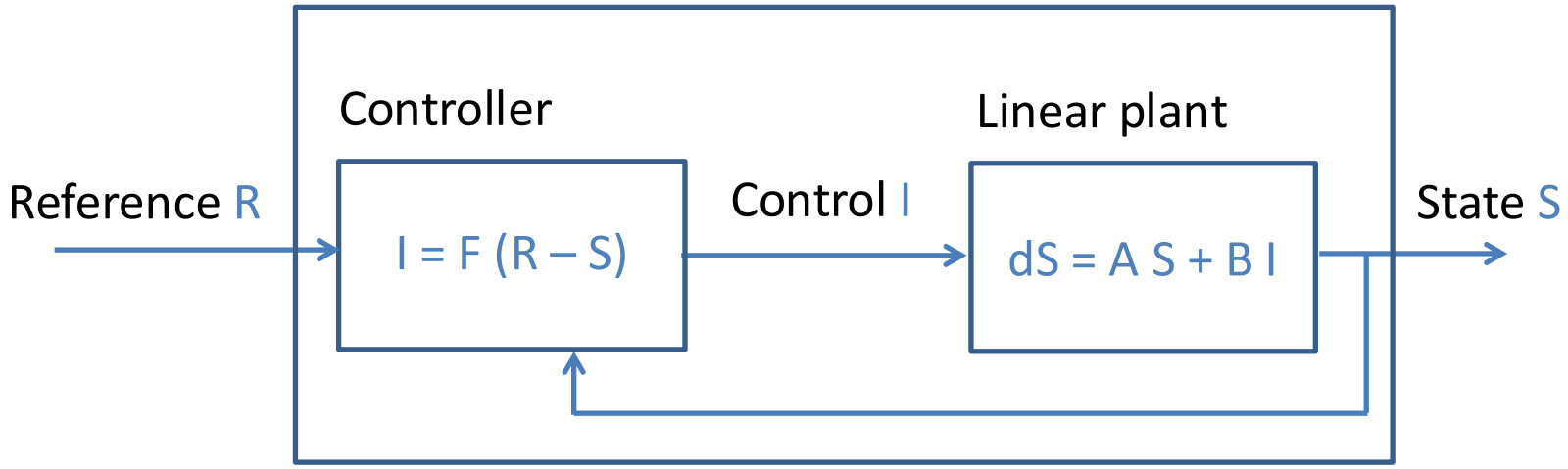
  - Rate of convergence depends on magnitude of K_P

Feedback Controller for Linear Systems



- Assume controller observes complete state vector S
- Reference signal R has same dimension as state vector S
- State feedback controller: linear transformation
- Matrix F: *gain matrix* of dimensions m×n, with m = |I|, n = |S|

Stabilization by Linear State Feedback



Composite system dynamics : $dS = (A - B\,F)\,S + B\,F\,R$

Goal of control design:

Define the gain matrix F so that the composed system is asymptotically, and so BIBO, stable

- Given A and B, find F such that each eigenvalue of $A - B\,F$ has negative real part

Design of Gain Matrix

System dynamics: $dS = A\,S + B\,I$ with n state and m input vars

Design goal: given A and B, find F such that each eigenvalue of $A - B\,F$ has negative real part

- When is this possible ?
- Suppose we choose desired eigenvalues $\lambda_1, \ldots, \lambda_n$ and solve the equation

$$\det(A - B\,F - \lambda I) = (\lambda - \lambda_1)(\lambda - \lambda_2) \ldots (\lambda - \lambda_n)$$

  where the $m \times n$ entries of matrix F are the unknowns

- When is this system guaranteed to be solvable?
- Does the existence of a solution depend on the choice of eigenvalues?

Controllability

Given an $n \times n$ matrix A and $n \times m$ matrix B, consider the *controllability* $n \times mn$ matrix

$$\mathbf{C}[A,B] = (B \quad AB \quad A^2B \quad \dots \quad A^{n-1}B)$$

m columns of B followed by m columns of $A\,B$, then of $A\,A\,B$, …

Recall: the *rank* of a matrix is the maximum number of linearly independent columns/rows

Definition: The matrix pair (A, B) is *controllable* if $\mathbf{C}[A,B]$ has rank n

Theorem: The following are equivalent:

1. The matrix pair (A, B) is controllable
2. For every set $\mathbf{\Lambda} = \{\lambda_1, \dots, \lambda_n\}$ of complex numbers s.t. $a + bj$ is in $\mathbf{\Lambda}$ iff its conjugate $a - bj$ is in $\mathbf{\Lambda}$,
   there is a $n \times m$ gain matrix F s.t. the eigenvalues of $A - B\,F$ are $\lambda_1, \dots, \lambda_n$

Example: Controllability test

Consider 2-dimensional system with one input u, with dynamics given by

$$d\,s_1 = 4\,s_1 + 6\,s_2 + 2\,u$$
$$d\,s_2 = s_1 + 3\,s_2 + u$$

- What are the matrices A, B, **C**[A, B]?
- What is the rank of **C**[A, B]?

Advantages of Controllability

Consider a linear system with dynamics:

$$dS = A\,S + B\,I \;\; ; \;\; \text{initial state } s_0$$

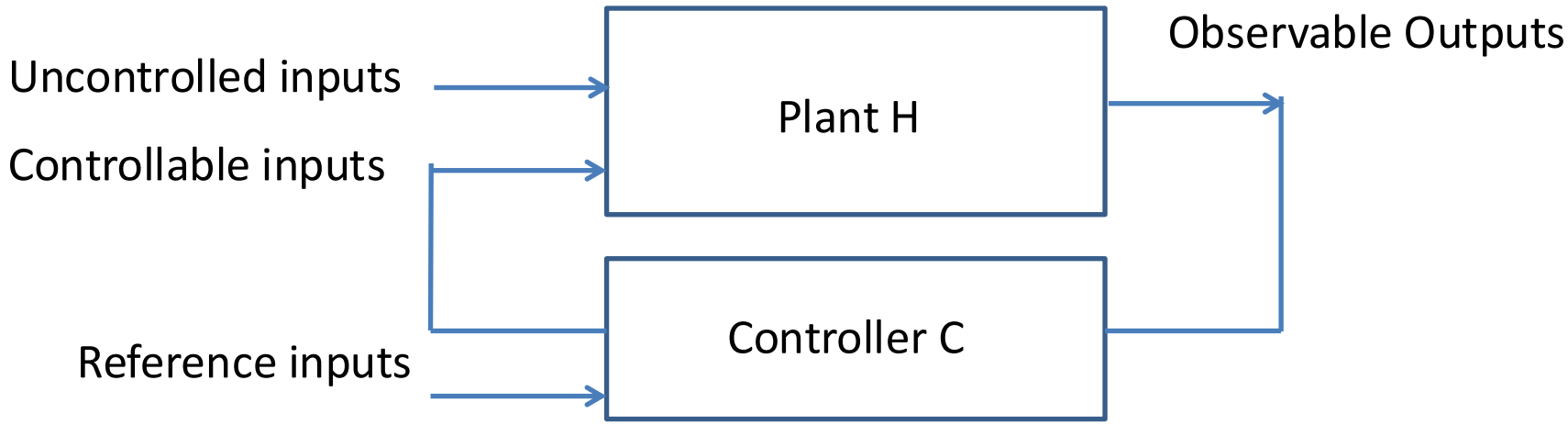
Suppose (A, B) is controllable

Then, for every system state s there is an input signal $\mathbf{I}$ and a time t_g such that

$$\mathbf{S}(t_g) = s$$

where $\mathbf{S}$ is the unique response signal for $\mathbf{I}$ and s_0

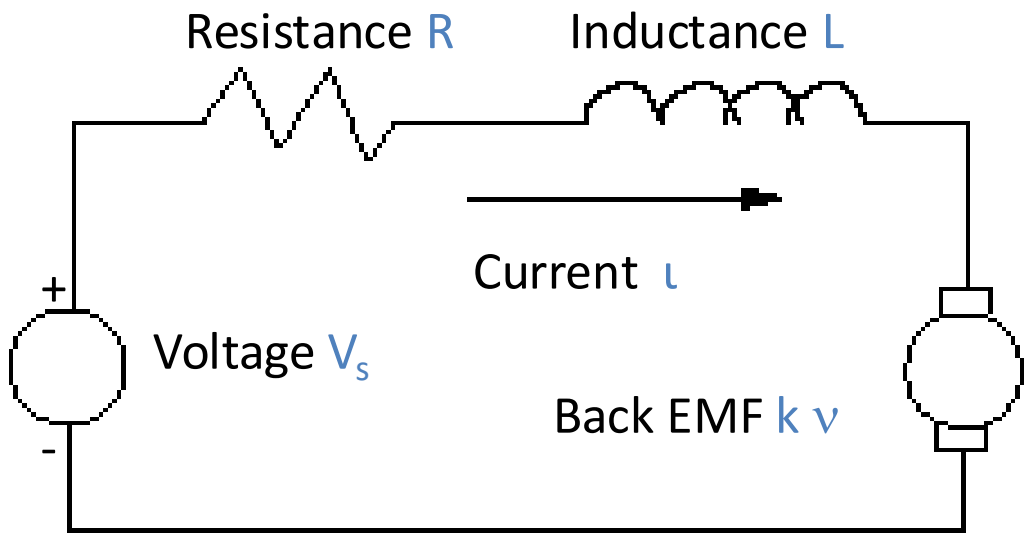
PID Controllers

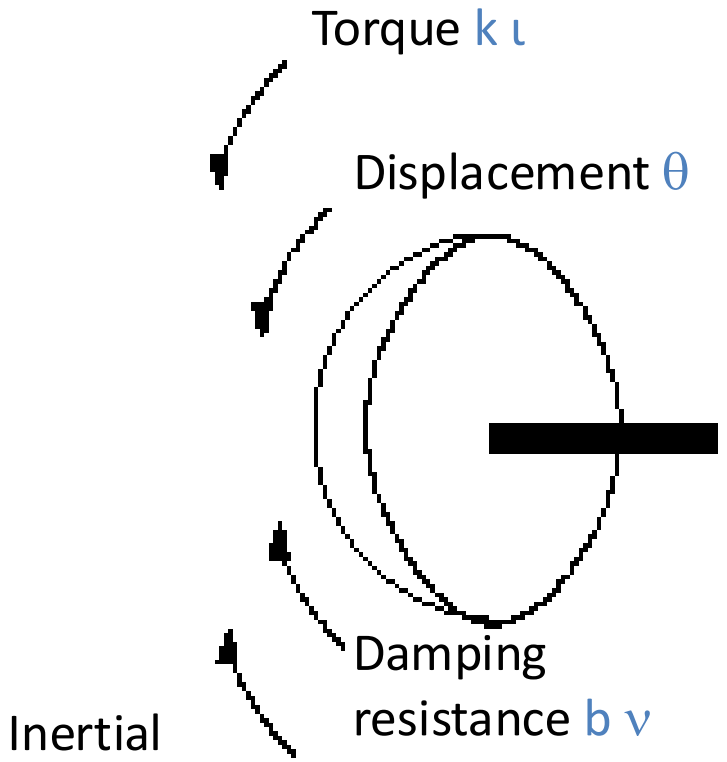


Widely used strategy for designing controllers in practice:

- Error = Reference Inputs – Observable Outputs
- Controller's output is sum of 3 terms:
  1. Term proportional to error
  2. Integral term to handle cumulative error
  3. Derivative term in response to error change rate

DC Motor





Laws of electrical circuits:

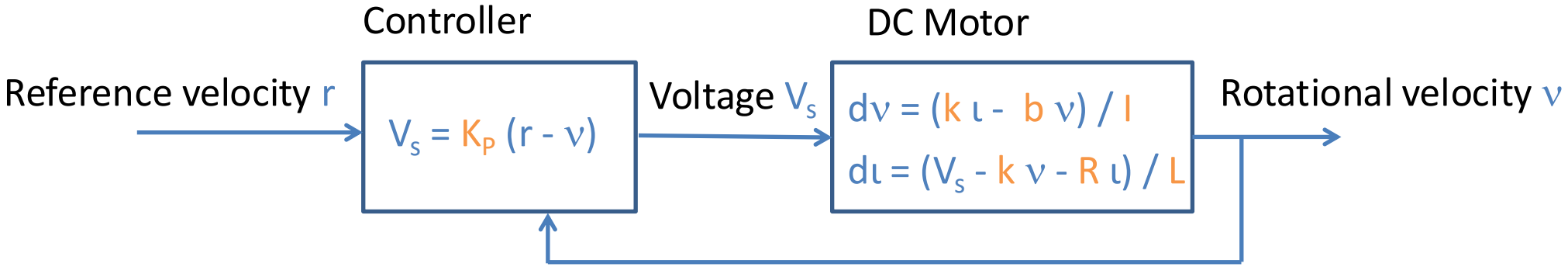
$\nu = d\theta/dt$

$V_s = L\, d\iota/dt + R\, \iota + k\, \nu$

Laws of motion for the shaft:

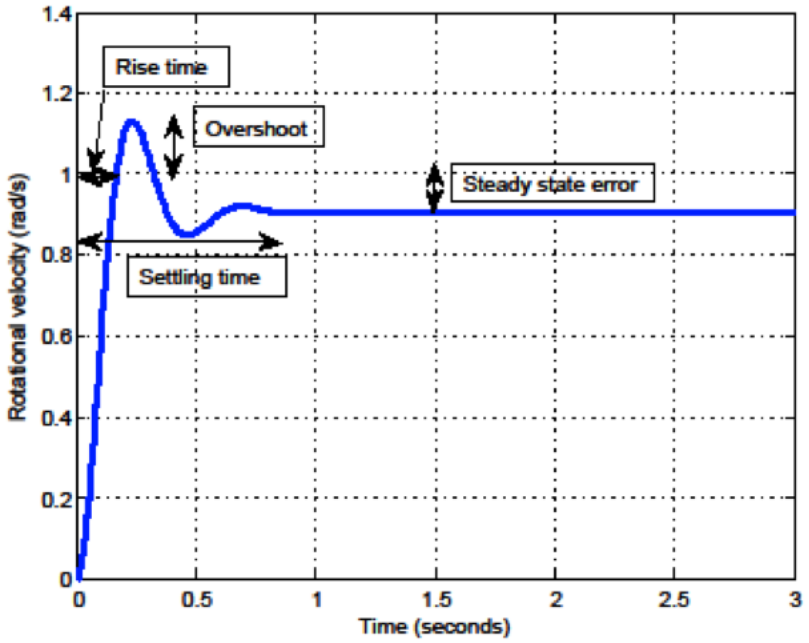
$I\, d\nu/dt + b\, \nu = k\, \iota$

Proportional Controller for DC Motor



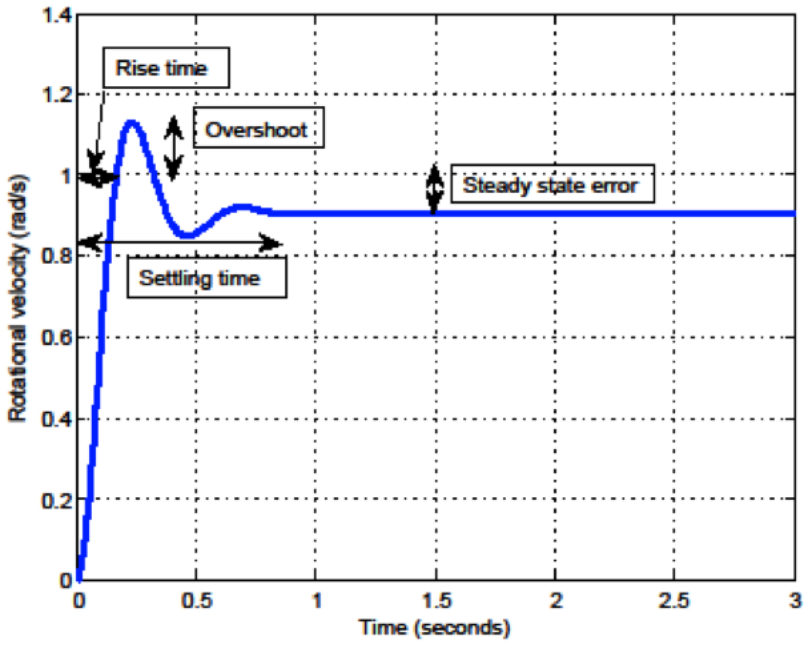
- DC Motor modeled as a linear system with
  - 2 state variables,
  - 1 input variable, and
  - 1 output variable
- Feedback controller observes rotational velocity ν, and adjusts voltage to make ν equal to desired velocity r
- First attempt: proportional controller (*P controller*)

Step Response of P Controller



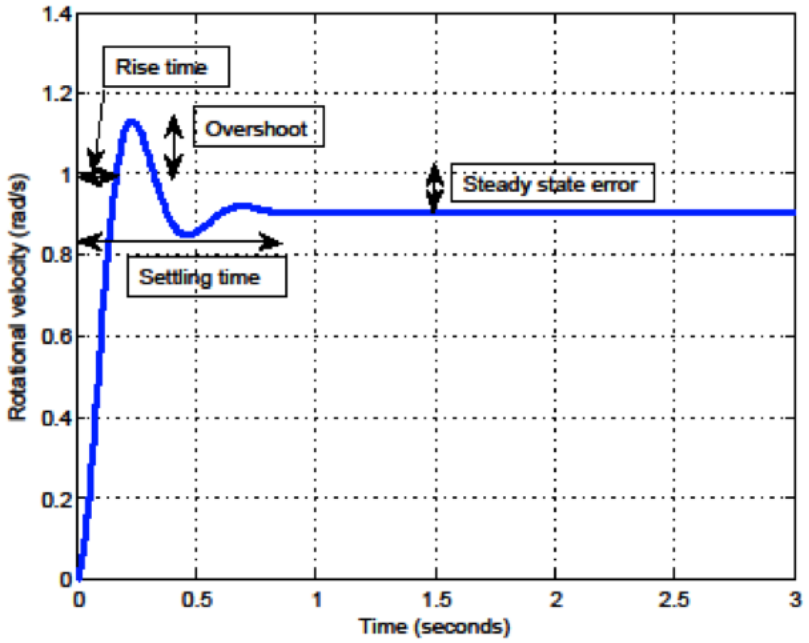
- **Step response:** How will system output change if at time 0, with $\nu = 0$, we change reference input r to 1?
- Plotted using MATLAB (see notes for values of various parameters)
- Beyond stability and convergence, what are desired characteristics of the response?

Characteristics of the Step Response



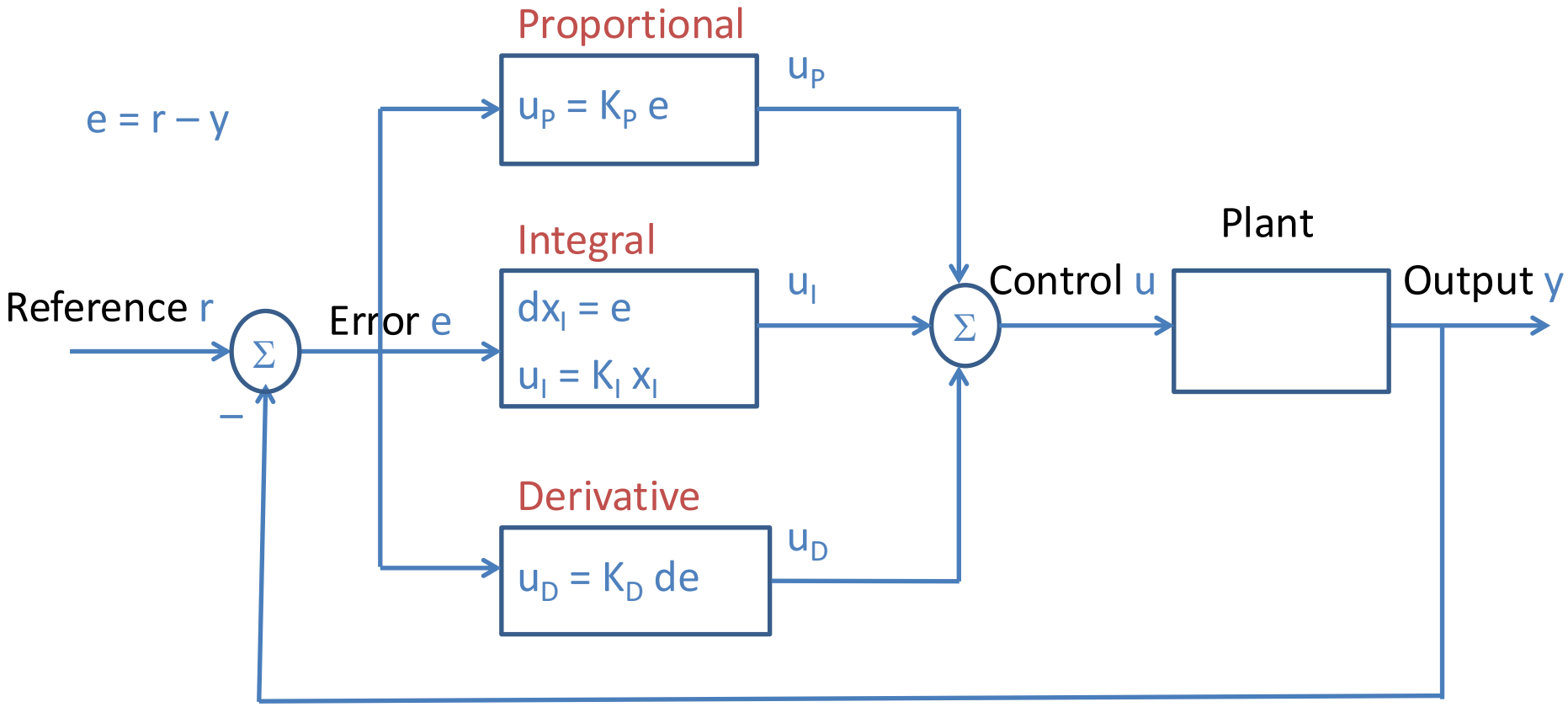
1. *Overshoot*: Difference between maximum output value and reference value (12% in this plot)
2. *Rise Time*: Time at which the output value crosses reference value (0.15sec in this plot)
3. Settling Time: Time at which output value reaches steady-state value (0.8sec in this plot)
4. *Steady State Error*: Difference between steady-state output value and reference (10% in this plot)

Improving the Step Response



- Performance of the P-controller depends on the value of the proportional gain constant K_P
- What happens if we increase it?
- Rise time decreases, but overshoot increases
- Steady-state error remains!
- Solution: Use integral and derivative gains

Generic PID Controller



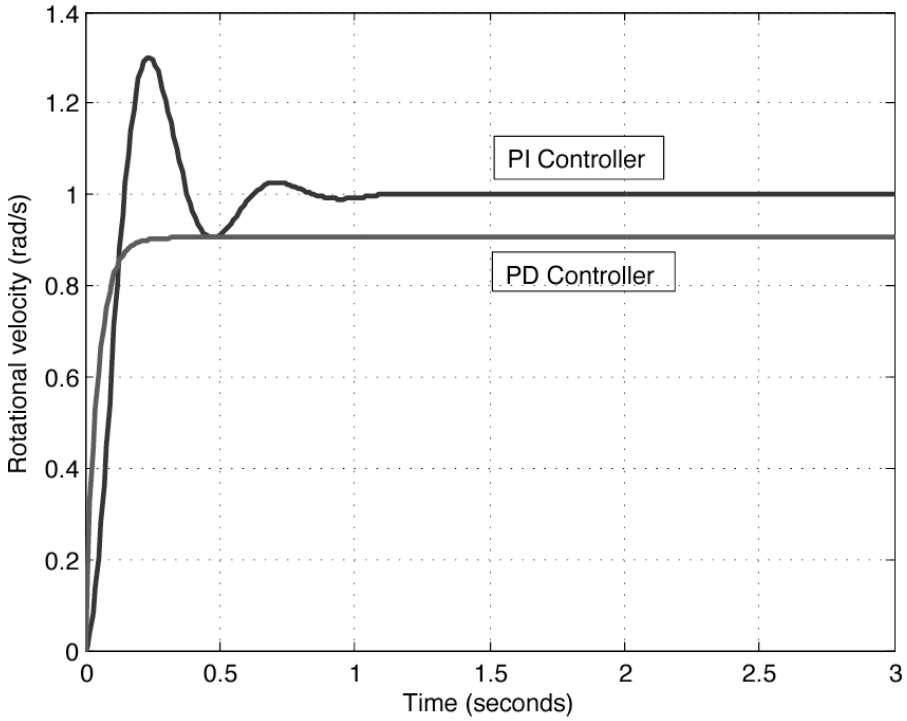
PID Controller

If **e**(t) is the error signal, then the output **u**(t) of the PID controller is sum of 3 terms:

1. **Proportional term:** K_P **e**(t), where K_P is the *proportional gain* (response to current error)
2. **Integral term:** $K_I \int_0^t$ **e**(t) dt, where K_I is the *integral gain* (response to error accumulated so far)
3. **Derivative term:** K_D (d/dt)**e**(t), where K_D is the *derivative gain* (response to current rate of change of error)

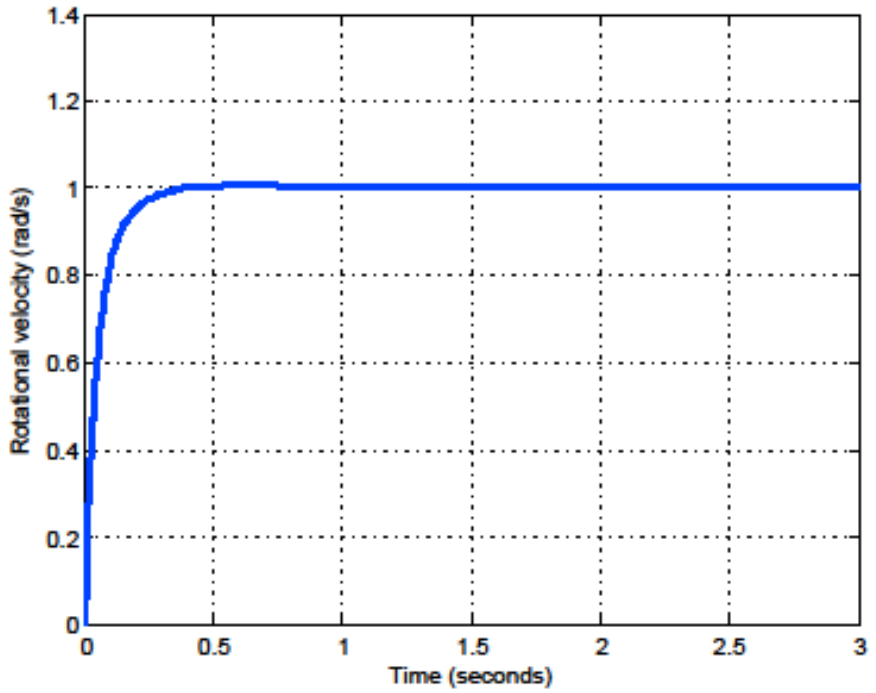
Controller special cases: P, PD, PI

PI and PD Controllers for DC Motor



- **PI controller:** adding integral term removes steady state error
  - Overshoot, rise time, setting time increase (why?)
- **PD controller:** adding derivative term removes overshoot
  - Steady state error remains

PID Controller for DC Motor



Excellent performance on all metrics: $K_P = 100$, $K_D = 10$, $K_I = 200$

Small rise time, settling time, negligible steady state error, no overshoot

Designing PID Controllers

What are the effects of changing the gain constants K_P, K_D, K_I ?

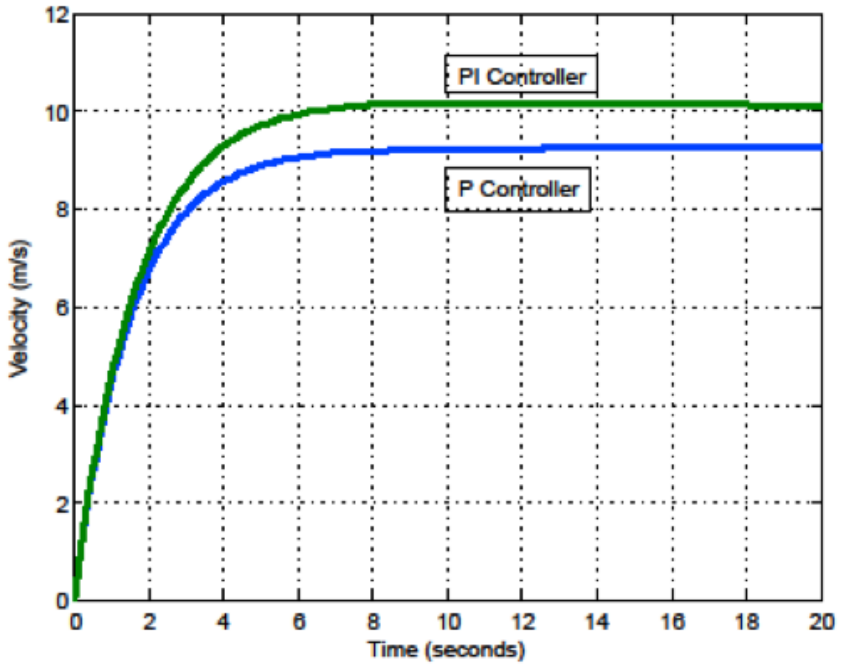
Broad co-relationships well understood

Control toolboxes allow automatic tuning of parameters

PID controllers seem to work well even when the actual system differs significantly from the plant model

- Computation of control output depends only on the measured error, and not on the model!

PI Cruise Controller



Desired velocity change: 10 m/s

PI controller: $K_P = 600$, $K_I = 40$

Settling time: 7s, with negligible overshoot and steady-state error

Works in real cars!

Credits

Notes based on Chapter 6 of

Principles of Cyber-Physical Systems
by Rajeev Alur
MIT Press, 2015